

ECE 310 — Wireless Communications Major Project

Milestone 1 (M1) Report

Robustness of Domain-Invariant CSI Motion Indicators under Synchronization Error and Bandwidth Scaling in Emerging IEEE 802.11bf Wi-Fi Sensing

Category 1: Research Project, Group 4 | Monsoon Semester 2026 | Milestone: M1 — Proposal & Scoping

Note on scope: As per guidance received from our project mentor, this report defines a clear problem direction/ subdomain selection, motivation, and preliminary technical grounding but not a final, locked problem statement. The specific research question will be sharpened at M2 alongside baseline reproduction, consistent with the course's evolving SOTA-anchor philosophy

1. Proposed Direction and Rationale

After comparing candidate directions within the Wi-Fi sensing generalizability landscape; Out of: Signal Preprocessing (signal analysis / motion indicators), Feature Learning (domain alignment, meta-learning, and related subdomains), and Model Deployment (transfer / federated / continual learning) we plan to work in the direction within Signal Preprocessing, specifically targeting the robustness of established domain-invariant motion indicators.

This decision was made on purpose even though there was an option to create a completely new way to measure movement that doesn't depend on the type of environment. That option was not chosen for this project because

(a) creating a motion indicator that works across all areas is the common topic in the whole field of Wi-Fi sensing generalization. There are methods that are already being used such as domain alignment, meta learning and disentanglement all trying to solve the same problem

(b) coming up with new signal processing math that is better than all of this in just one semester is very risky, for a team that is still learning the basics. Thus we instead decided to carefully look at a method that is already trusted and used a lot and plan tests it in situations where it was never checked before, this seems to be a safer approach that starts with proof and still has a real chance to improve the way things are done (a specific fix) once the testing is done.

2. Problem Area and Motivation

2.1 Background

Wi-Fi sensing works by using Channel State Information (CSI). CSI is a record of the amplitude and phase for each subcarrier. This data is already produced by Wi-Fi hardware when it corrects the signal for equalisation. Wi-Fi sensing uses this data to spot people, their gestures and their actions all without cameras or wearables. A big problem for putting Wi-Fi sensing into life is generalisation. Models that are trained in one room on one device or with one user usually lose a lot of accuracy when any of those conditions is different.

To address this, the field has moved towards domain-invariant motion indicators. The common ones are Body-coordinate Velocity Profile (BVP) and Doppler Frequency Shift (DFS). BVP and DFS pull out physics-based motion features, such as how fast body parts move or the frequency components that shift because of motion, which are, in principle, independent of the specific room or device. These indicators, introduced in Widar 3.0 [1], are treated by most subsequent generalization research as a stable, reliable preprocessing stage on top of which domain-alignment and adaptation algorithms are built.

2.2 Why This Assumption Is Worth Re-Examining Now

Raw CSI measured on any commodity Wi-Fi receiver is already known to be corrupted by hardware-induced phase artifacts even if no deliberate change is made: Carrier Frequency Offset (CFO), Carrier Phase Offset / Phase-Locked Loop offset (CPO), Sampling Frequency Offset (SFO), and Packet Detection Delay (PDD) [5], [6], [7]. These problems appear because the transmitter clock and the receiver clock are never exactly synchronized. Prior work has partially corrected these problems for general CSI-based sensing tasks (ranging or respiration monitoring) [5], [6]. However, the specific impact on BVP/DFS-style domain-invariant motion indicators – the layer that lies directly under cross-domain generalization research – has not yet been studied in detail and understanding CSI under these conditions remains a key challenge.

Two concurrent developments make this gap newly important:

- Standardization of IEEE 802.11bf: the WLAN Sensing amendment, finalized in 2024 defines dedicated sensing procedures across sub-7 GHz and mmWave bands and enables multiband operation [2], [3]. Recent work on multiband passive sensing explicitly flags synchronization (AP-clock asynchrony) across bands as an open, unresolved implementation problem [4].
- Bandwidth and hardware diversification: sensing-capable hardware is moving toward wider bandwidths and more diverse receiver chipsets than the narrowband (20/40 MHz), single-vendor (e.g., Intel 5300) conditions under which BVP/DFS were originally validated. A 2026 cross-receiver study found that receiver-induced variability (dominated by Automatic Gain Control and subcarrier non-linearities, not timing) already measurably degrades cross-device Human Activity Recognition accuracy, recoverable up to 75% with a gain-alignment preprocessing step [8]. That same study explicitly separates gain effects from timing/synchronization effects and does not characterize the latter for domain-invariant velocity-based indicators specifically which is the gap we intend to target.

This project's direction is therefore precisely scoped: quantify how BVP/DFS-style indicators behave as

(i) synchronization error (timing/CFO-driven phase drift)

(ii) bandwidth/subcarrier count are varied toward conditions relevant to 802.11bf-era deployment, using existing public CSI data and controlled, simulated perturbation rather than new hardware

2.3 Significance

- Practical: if BVP/DFS degrade sharply under realistic timing/bandwidth variation every downstream generalization technique built on top of BVP/DFS (domain alignment, meta-learning, transfer learning) inherits a fragility hidden from accuracy benchmarks run on today's narrowband datasets.
- Wireless-communications relevance: the project is anchored in core ECE 310 concepts. Carrier/timing synchronization, bandwidth, subcarrier structure and the resulting SNR/phase-coherence degradation rather than being a purely a machine-learning exercise.

- Timeliness: 802.11bf was finalized in 2024, and both standards-affiliated groups (NIST) and academic groups are actively characterizing its sensing procedures and hardware effects [2], [3], [8]; a signal-level robustness study of existing trusted indicators complements this emerging body of work rather than competing in an already saturated corner of the field.

2.4 Assumptions and Scope (initial, to be refined at M2)

- We assume existing public CSI datasets (Widar 3.0 [1]) gives us raw CSI data so that we can create controlled signal-level changes like adding timing offsets or reducing bandwidth/subcarrier resolution without without new hardware data collection.
- We limit the study to a single domain-invariant indicator, BVP and a single public dataset Widar 3.0. We will consider datasets or indicators such as DFS only if time allows.
- At this stage we are not fixing a correction method. Whether we propose a correction and what it looks like will depend on what the M2/M3 sensitivity analysis shows. We wil follow our mentors advice to let the problem statement become clearer as evidence comes in than setting a solution too early.
- Out of scope for this semester: real 802.11bf hardware deployment; mmWave/multiband fusion; full integration with federated or continual learning pipelines.

3. Review of Existing Work

The following table summarizes the literature most directly relevant to our direction, organized to make explicit what each work assumes, measures, and leaves open.

#	Paper	Problem / Focus	Method	Key Assumption	Metric	Limitation Relevant to Us
1	Zheng et al., MobiSys 2019 (Widar 3.0) [1]	Cross-domain gesture recognition without retraining	Extracts BVP, a domain-invariant velocity feature, from raw CSI	Narrowband (20 MHz), tightly synchronized commodity hardware	82.6-92.4% cross-domain accuracy	Never tests robustness to timing offset or bandwidth change
2	Ropitault et al., IEEE Commun. Std. Mag. 2024 [2]	Standardizing WLAN sensing procedures across devices/bands	Defines 802.11bf initiator/responder and multiband sensing roles	Standards-document scope	-	Establishes multiband/wide-bandwidth sensing as an imminent reality; motivates why our stress test matters
3	Sahoo et al., arXiv:2403.19825, 2024 [3]	Quantifying 802.11bf sensing overhead in sub-7 GHz band	Simulation-based performance evaluation across configurations	Standard-compliant sensing procedures	Overhead / accuracy trade-off	Evaluates protocol overhead, not how existing ML indicators behave under resulting timing/bandwidth conditions
4	Picazo-Martinez et al., arXiv:2507.22591, 2025 [4]	Reusing multiband Wi-Fi signaling for passive sensing	System-level multiband (sub-7GHz + mmWave) sensing design	Multiband nodes approximately synchronized	Feasibility discussion	Explicitly flags AP-clock asynchrony as unresolved - direct justification for our independent variable
5	Ratnam et al., IEEE Trans. Wireless Commun. 2024 [5]	General CSI gain/phase error correction for sensing	Mathematical model of gain error and synchronization-based phase error; correction algorithms	Errors are separable from true channel variation via modeling	SNR improvement (20-200%)	Corrects general CSI for tasks like respiration monitoring; does not evaluate motion-indicator (BVP/DFS) robustness specifically
6	Zubow, Gawlowicz, Dressler, WONS 2021	Empirical characterization of 802.11ac phase offsets	Measurement study of commodity Wi-Fi phase-offset behavior	COTS 802.11ac hardware	Phase offset statistics	Characterizes the raw phenomenon, not its effect on downstream generalization indicators
7	Gringoli et al., arXiv:2605.26836, 2026 [8]	Cross-receiver variability in CSI for HAR	Systematic multi-receiver comparison; gain-alignment preprocessing	Variability is dominated by AGC/gain and subcarrier non-linearity	HAR accuracy recovery (up to 75%)	Finds gain/AGC dominates for HAR robustness; explicitly separates out and does not deeply characterize timing/synchronization effects on velocity-based indicators - the gap we target
8	Li et al., WiTTA-Bench, CVPR 2026	Benchmarking test-time adaptation	Evaluates 20 TTA methods across environment/subject/device shift protocols	Domain shift is corrected at the algorithm level, post-	Cross-domain accuracy under shift	Operates one layer above signal preprocessing; does not examine signal-level

		under domain shift		feature-extraction		(timing/bandwidth) causes of shift
9	Wang et al., IEEE Commun. Surveys Tuts. 2026 [9]	Mapping the Wi-Fi sensing generalization research landscape	Four-stage taxonomy: Experimental Setup, Signal Preprocessing, Feature Learning, Model Deployment	-	-	Frames BVP/DFS as largely settled tools; does not flag bandwidth/synchronization robustness as an examined sub-question

3.1 Synthesis of the Review

Two distinct research communities are currently working on this problem without meeting in the middle. The Wi-Fi sensing generalization community ([1], [9]) views BVP and DFS as reliable components. They focus on building advanced domain-alignment and adaptation methods that rely on these signals. Separately, the signal-processing / standards community ([2]-[6], [8]) studies raw CSI data and actively identifies and partially fixes hardware-related issues like timing errors and gain variations. However their work evaluates how these artifacts affect sensing tasks (ranging, respiration monitoring, raw HAR accuracy) rather than on the specific domain-invariant motion indicators that generalization research depends on.

The closest existing work, Gringoli et al. [8] (2026), is an important reference point, that differences in CSI between receivers are mostly due to gain and automatic gain control (AGC) effects than timing issues when it comes to standard HAR. It also finds that a simple gain alignment step can recover most of the accuracy. This is valuable evidence but leaves two questions explicitly open that this project targets:

- (i) whether that finding still holds specifically for BVP/DFS-style velocity indicators (rather than generic HAR pipelines) as timing/synchronization error is deliberately varied across a wider range motivated by 802.11bf multiband conditions
- (ii) how robust these indicators behaves as bandwidth and subcarrier count scale toward 802.11bf-era configurations, which was outside that study's scope. To date no paper in this review links indicator robustness directly to both factors: timing and scaling, under world standards-driven perturbations.

4. Initial State-of-the-Art (SOTA) Position

Primary SOTA target: Zheng et al.'s BVP-based Widar 3.0 framework [1] remains the most cited and reused domain-invariant motion indicator in the field and is the direct focus of our study.

Closest related SOTA (differentiator required): Gringoli et al. [8] (2026) and Ratnam et al. [5] (2024) already study CSI hardware-induced variability and correction, but at the level of general CSI/HAR robustness, not at the level of BVP/DFS-style domain-invariant indicators under 802.11bf-motivated timing and bandwidth ranges.

What SOTA does that we do not (yet): [1] achieves strong, validated cross-domain (room/user/orientation) accuracy under its originally tested narrowband, synchronized conditions. [5], [8] already provide correction techniques for general CSI gain/phase artifacts.

What we aim to do that SOTA does not: characterize how BVP/DFS specifically behaves as timing synchronization error and bandwidth/subcarrier count are varied across 802.11bf-motivated ranges, and We also want to find out whether existing general-purpose corrections, (e.g., [5]'s phase correction, [8]'s gain alignment) are sufficient for this indicator or whether a targeted correction is needed.

This position is provisional and will be re-evaluated at M2 once baseline reproduction of the BVP pipeline is complete, as per the course's M1-M4 evolving SOTA-anchor philosophy

5. Timeline (Direction-Level)

Weeks	Milestone	Activity
1-3 (M1)	Direction scoping	Literature review (this report); confirm Widar 3.0 raw CSI/BVP access for perturbation experiments; finalize perturbation variable ranges from standards literature [2],[3],[5]
4-5 (M2 prep)	Baseline reproduction	Reproduce Zheng et al.'s BVP pipeline on Widar 3.0 at nominal (unperturbed) conditions; refine research question and problem statement based on reproduction results, per mentor's guidance
6-7	Technical artifact	Implement controlled injection of timing offset (CFO/CPO-style phase drift) and bandwidth/subcarrier reduction into existing CSI data
8-9	First evaluation	Run BVP classifier under perturbed conditions; obtain initial sensitivity curve; compare against Gringoli et al. [8]-style gain-alignment and Ratnam et al. [5]-style phase-correction as candidate baselines
10-12	Extensive experiments	Full sweep across offset/bandwidth ranges; identify failure threshold; explore targeted correction if warranted by findings
13-14	Documentation	Final report, reproducibility package
15	Final presentation	- And continue work for phase 2

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